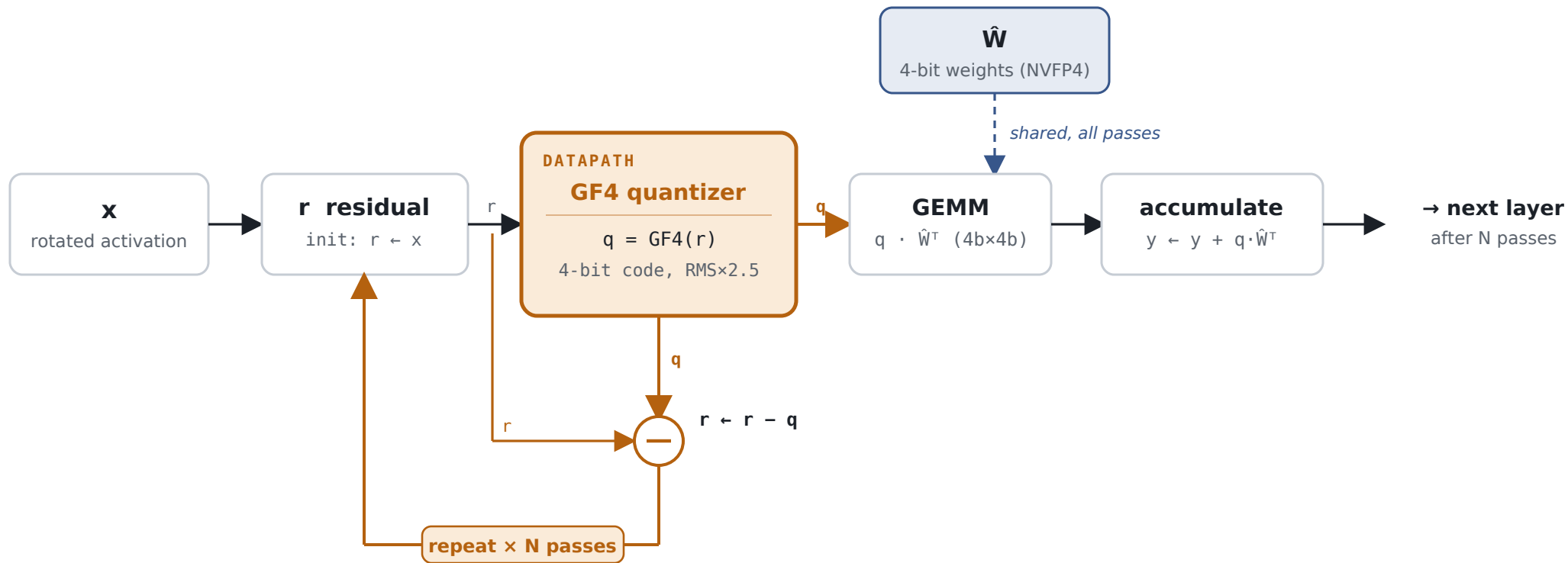


MULTI-PASS RESIDUAL · QUANTIZED FORWARD PASS



$$q_1 + q_2 + \dots + q_N \approx x \Rightarrow y = (q_1 + \dots + q_N) \cdot \hat{W}^T \approx x \cdot W^T$$

one 4-bit GEMM, reused N times — the residual (the error) is re-sent through the same weights: no second high-precision datapath.